

- – Ensure enabling system or service access, and materials, needed to support implementation.
- – Schedule the Implementation process, noting any critical path elements and ensuring that those critical element issues/constraints are addressed and understood by all stakeholders.
- *Perform implementation.*
 - – Realize the system elements per the detailed product, process, and material specifications.
 - – Produce documented evidence of implementation compliance:
 - – Complete detailed product, process, material specifications, and system configurations.
 - – Conduct peer reviews and testing—Inspect and verify software and hardware for correct functionality, hardware functional testing, etc.
 - – Conduct conformation audits—Compare hardware and software elements to detailed drawings and design artifacts to ensure that each element meets its detailed specifications prior to integration with other elements.
 - – Prepare initial training capability and draft training documentation—To provide the user community with the ability to operate, conduct failure detection and isolation, conduct contingency scenarios, and maintain the system as appropriate.
 - – Prepare a hazardous materials log, if applicable.
- *Manage results of implementation.*
 - – Identify and record implementation results. Provide baseline information for configuration management (see [Section 2.3.4.5](#)). Maintain the records per organizational policy (see Section 2.2.3.6).
 - – Record any anomalies encountered during the Implementation process and resolve the anomalies (corrective actions or improvements) using the Quality Assurance process. (see [Section 2.3.4.8](#))
 - – Establish and sustain traceability of the implemented system elements with the system architecture, design, and system and interface requirements that are needed for implementation (see [Section 3.2.3](#)).

Common approaches and tips.

- Nearly all implementations have some issues requiring adjustments. Schedule pressures to get the system into operation can lead to lapses in tracking the needed adjustments. To assist consider the following:
- Keep the team (e.g., the Integrated Product Development Team (IPDT)) engaged to assist with configuration issues and redesign.
- Inspections are a proactive way to build in quality (Gilb and Graham, 1993).
- Conduct hardware conformation audits or system element level hardware verification; and ensure sufficient software configuration verification prior to entering the Transition process.
- Identify the enabling systems and materials needed for implementation early in the life cycle to allow for the necessary lead time to obtain or access them.

Elaboration

Implementation Concepts. The implementation process typically focuses on the following four forms of system elements:

- Hardware—Output includes fabricated or adapted physical elements.
- Software—Output includes software code and executable images
- Operational resources—Output includes procedures and training.
- Services—Output includes specified services. These may be the result of one or more hardware, software, or operational elements resulting in the service.

The Implementation process can support either the creation (fabrication or development) or adaptation of system elements. For system elements that are reused or acquired (such as COTS), the Implementation process allows for adaption of the elements, if necessary, to satisfy the needs of the SoI. This may be accomplished via configuration settings provided within the element (e.g., hardware configuration switches and software configuration tables). Newly created products have more flexibility to be designed and

developed to meet the needs of the SoI without modification (but at increased cost).

2.3.5.8 Integration Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.4.8.1] The purpose of the Integration process is to synthesize a set of system elements into a realized system that satisfies the system requirements.

Description The focus of integration is the combination of system elements (hardware, software, and operational resources) that compose the SoI and verifying the correctness of the static and dynamic aspects of interfaces between, and interaction among, the implemented system elements. Integration also includes proactive activities to address potential integration issues early in the project such as modeling, analysis, simulation, prototyping, and early testing. Integration constraints and objectives are identified and considered during the definition of the system requirements, architecture, and design. The interaction of the Integration process with the system definition processes (i.e., System Requirements Definition, System Architecture Definition, and Design Definition) early in the development stage is essential for avoiding integration issues during the system realization.

The Integration process works closely with the Verification and Validation (V&V) processes. This process is iterated with the V&V processes, as appropriate, and includes an assessment of the integration maturity of elements to be integrated. As the integration of system elements occurs, the Verification process is invoked to check the correct implementation of system requirements, architectural characteristics, and design properties. The Validation process may be invoked to check that the individual system elements meet the stakeholder requirements and provide the function

intended. The process checks that all boundaries between system elements have been correctly identified and described, including physical, logical, and human–system interfaces and interactions (physical, functional, sensory, and cognitive), and that all system and system element functional and performance requirements and constraints are satisfied.

Inputs/Outputs Inputs and outputs for the Integration process are listed in [Figure 2.48](#). Descriptions of each input and output are provided in Appendix E.

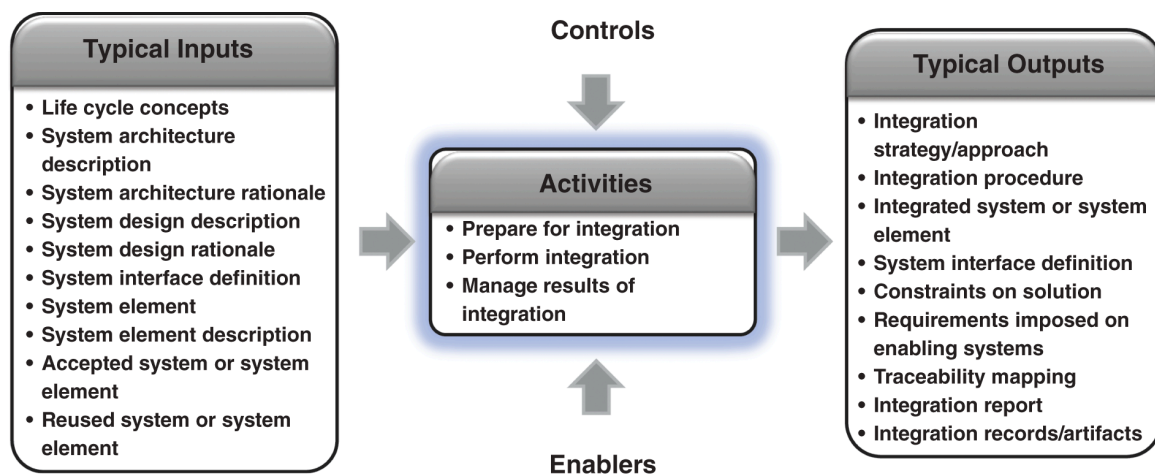


FIGURE 2.48 IPO diagram for Integration process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Integration process includes the following activities:

- *Prepare for integration.*
 - – Establish checkpoints for the correct implementation of the interfaces as the system elements are progressively aggregated.
 - – Establish the strategy/approach for integration.
 - – The strategy identifies risk mitigation approaches and sequences the order and levels for aggregating system elements while considering integration time and cost.
 - – Identify system constraints and objectives necessary for successful integration to be addressed in the system requirements, architecture, or design such as those for accessibility, safety for integrators, required interfaces for enablers.

- – Plan for the necessary enabling systems or services needed to support integration such as integration facilities, training systems, or simulators.
- – Ensure enabling system or service access, and materials, needed to support integration.
- *Perform integration.*
 - – Integrate system element configurations until the system is complete.
 - – Manage interface availability as scheduled and track conformance of the interfaces to their requirements.
 - – Address any conformance or availability issues.
 - – Integrate the system elements in accordance with planned sequences.
 - – Perform check of the interfaces, selected functions, and critical quality characteristics at different integration levels.
- *Manage results of integration.*
 - – Capture the integration results, including any anomalies or other issues identified. This includes anomalies due to the integration strategy, the integration enabling systems, execution of the integration, or incorrect system or element definition.
 - – Where inconsistencies exist at the interface between the system, its specified operational environment, and any systems that enable the utilization stage the deviations lead to corrective actions or requirement changes. The Project Assessment and Control process (see [Section 2.3.4.2](#)) is used to analyze the data to identify the root cause, direct corrective or improvement actions, and to record lessons learned.
 - – Maintain bidirectional traceability of the integrated system elements and the strategy, plans, and requirements (see [Section 3.2.3](#)).
 - – Give CM the information items, work products, or other artifacts needed for baselines. The Configuration Management process (see [Section 2.3.4.5](#)) is used to establish and maintain baselines.

Common approaches and tips:

- The integration strategy should account for the schedule of availability of system elements and account for the personnel that will use, operate, maintain, and sustain the system). It should also be consistent with the defect/fault isolation and diagnosis practices.
- Development of integration enablers, such as tools and facilities, can take as long as the system itself and should be started early in the project.
- The Integration process of complex systems should use flexible approaches and techniques.
- Integrate aggregates in order to detect faults more easily. The use of the coupling matrix technique applies for all strategies and especially for the bottom-up integration strategy (see [Section 3.2.4](#)).

Elaboration

Integration occurs throughout the project from initial needs identification through utilization and support. The focus of integration evolves as the system evolves from concept definition to system definition to system realization to system deployment and use. As the system progresses, the emphasis of integration changes from its system definition, analysis, modeling, or prototypes to the deployed and operational system integrated into its intended environment, including interfacing systems. Integration should look proactively to mitigate risks and avoid integration issues, or discover them at the earliest point.

Concept of an “Aggregate.” The integration of a system is based on the notion of an “aggregate.” An aggregate is made up of several system elements and their physical and functional interfaces. Each aggregate is characterized by a configuration that specifies the system elements to be integrated and their configuration status. A set of verification actions is applied on each aggregate. To perform these verification actions, a verification configuration that includes the aggregate plus verification enabling systems is constituted. The verification enabling systems can be simulators (simulated system elements), emulators, stubs or caps,

scaffolding, activators (launchers, drivers), harnesses, measuring devices, etc.

Integration Strategy and Approaches. The integration of evolving system elements is performed according to a predefined strategy. The strategy relies on the defined physical and functional architectures of the system and the organizational structure developing it. The detailed implementation of the strategy is described in an integration plan that defines the actions to be taken to mitigate integration risks and the configuration of expected aggregates of evolving system elements. It also defines the sequence these aggregates to carry out efficient verification actions and validation actions (e.g., inspections, analyses, demonstrations, or tests). The integration strategy is thus elaborated in coordination with the selected verification strategy and validation strategy (see [Sections 2.3.5.9](#) and [2.3.5.11](#)).

Several possible integration approaches and techniques can be used to define an integration strategy. Any of these may be used individually or in combination. The selection of integration approaches and techniques depends on several factors, in particular the type of system element, delivery time, order of delivery of system elements, risks, constraints, etc. Each integration approach has strengths and weaknesses, which should be considered in the context of the SoI.

Integration of the SoI and enabling systems occurs during development as well as utilization and support. Early in the life cycle, integration is concerned with concepts, requirements, architecture, and design. Approaches include models, analysis, simulations, and prototypes. In later life cycle stages, integration focuses on changes during utilization and support.

There are multiple options for the combination of system elements or aggregation of completed system elements or aggregates. Some common integration techniques are:

- *Global (or Big Bang) integration*—The simplest approach for low-risk, complicated, or simple systems is integration of the entire SoI. While the process is simplified, any issues or interface problems are difficult to find and resolve.
- *Bottom-up integration*—A common approach follows the reverse order of decomposition from lowest system element through levels of the architecture to the final system. Problems can be found at lower levels and more easily isolated to specific system elements. System level issues may not be discovered until late in the process.
- *Top-down integration*—This is a common variation of incremental integration (see below) that starts with the system elements that most closely reflect overall system performance with peripheral elements simulated and integrated later. The purpose is to detect system level issues, particularly with external interfaces, early.
- *Incremental integration*—In a predefined order, one or a small number of system elements are added to an already integrated increment of system elements. It can also include a portion of the system being integrated into a predefined increment. This approach can be effective for incremental and evolutionary development (see [Section 2.2](#)). For agile development, the order can be defined by features.
- *Subset integration*—System elements are assembled by subsets, and then subsets are assembled together. Subsets can be defined by functional chains or threads to perform specific tasks.
- *Criterion-driven integration*—The most critical system elements compared to the selected criterion are first integrated (e.g., dependability, complexity, technological innovation). The criteria are generally related to risks. This technique allows early integration and verification of intensively critical system elements.
- *Integration “with the stream”*—The delivered system elements are assembled as they become available.
- *Model-based integration*—The system elements are modeled physically or functionally and integrated in the model environment. Actual system elements can be inserted into the model environment as they developed.

Throughout the project, the integration strategy addresses management approaches to address risks such as communications issues. These include use of Integrated Product Teams (IPTs), Interface Control Working Groups (ICWGs), Systems Engineering Integration Teams (SEITs), or Technical Performance Measures (TPMs).

Horizontal & Vertical Integration. The Integration process needs to address the wide range of integration perspectives that apply across the life cycle. Horizontal integration typically refers to activities that are performed across elements that appear in a common hierarchy level of the system architecture. Structural aspects may be system elements that collectively constitute a system. Behavioral aspects include the sequence of discrete behaviors that together describe system functionality. Vertical integration typically refers to activities that are performed to help ensure that system elements at a given system hierarchy level are consistent with, and satisfy the expectations of, the system or higher-level system elements. The recursive nature of SE highlights how integration features span the levels of the system structure (see [Section 2.3.1.2](#)). As there is new information or learning on one level of the system structure, it is shared with both higher and lower levels. Other integration “directions” span additional viewpoints and stakeholder concerns, such as those relating to temporal or functional considerations, application of standards, satisfaction of regulatory expectations, or operational conditions and environments. Integration can also be viewed in relationship to the requirements concepts of horizontal traceability among parallel elements in the architecture and vertical traceability between system hierarchy levels (see [Section 3.2.3](#)).

2.3.5.9 Verification Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.4.9.1] The purpose of the Verification process is to provide objective evidence that a system, system element, or artifact fulfils its specified requirements and characteristics.

Description The Verification process can be applied to any engineering artifact, entity, or information item that has contributed to the definition and realization of the SoI (e.g., verification of stakeholder needs, stakeholder requirements, system requirements, models, simulations, the system architecture, design characteristics, verification procedures, or a realized system or system element). The Verification process provides objective evidence with an acceptable degree of confidence to confirm:

1. The artifact or entity has been made “right” according to its specified requirements and characteristics,
2. No anomaly (error/defect/fault) has been introduced at the time of any transformation of inputs into outputs.
3. The selected verification strategy, method, and procedures will yield appropriate evidence that if an anomaly were introduced, it would be detected.

As is often stated, verification is intended to ensure that “the artifact or entity has been built right,” while validation is intended to ensure that “the right artifact or entity will be or was built.”

Inputs/Outputs Inputs and outputs for the Verification process are listed in [Figure 2.49](#). Descriptions of each input and output are provided in Appendix E.

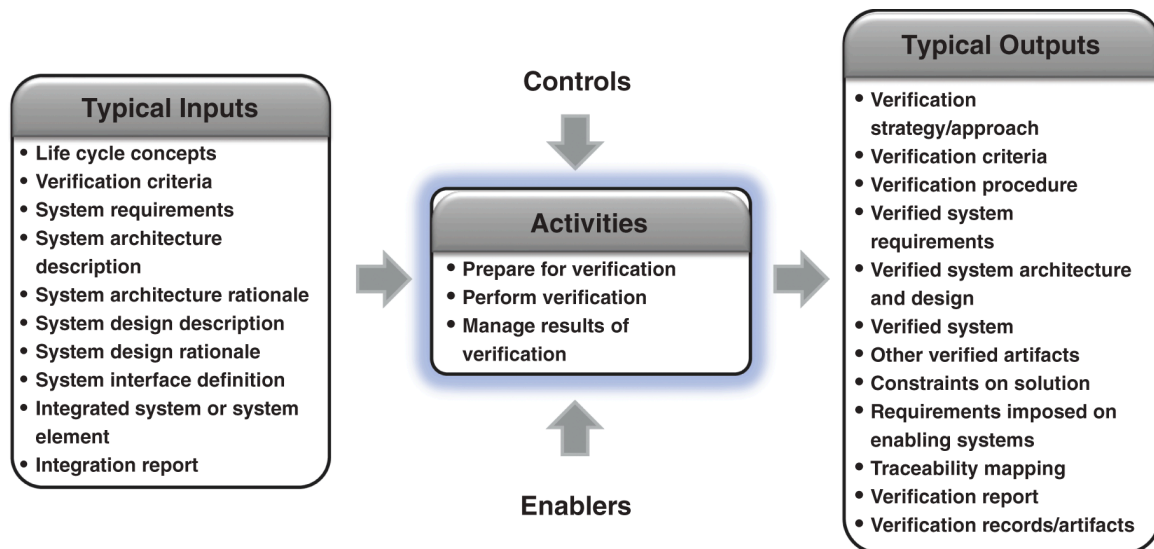


FIGURE 2.49 IPO diagram for Verification process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Verification process includes the following activities:

- *Prepare for verification.*
 - – Define the scope (what will be verified) and the verification actions (strategy, method, and success criteria). Verification activities consume resources: time, labor, facilities, and funds. The scope of the organization’s verification strategy/approach should be documented within the project’s SEMP and system integration, verification, and validation plans.
 - Establish a list of entities to be verified, including stakeholder needs, stakeholder requirements, system requirements, system architecture, prototypes, models, simulations, the system design, design characteristics, the system elements within the SoI architecture, and the integrated SoI itself.
 - Identify the specified requirements against which each entity will be verified.
- – Consider and capture constraints that could impact the feasibility or effectiveness of verification actions. The constraints could impact the implementation of the verification actions and include contractual constraints, limitations due to regulatory requirements, cost, schedule, feasibility to exercise a function, safety and security

considerations, the laws of physics, physical configurations, accessibility, etc.

- – For each verification action, select one or more verification methods and associated success criteria. Verification methods include inspection, analysis, demonstration, and test (each of these methods are defined later in this section). The success criteria define what the verification actions must do that will result in sufficient objective evidence to show that the entity has fulfilled the requirement(s) against which it was verified against.
- – Establish the strategy/approach for verification, including trade-offs between scope and constraints. The verification strategy includes the method that will result in objective evidence that the verification success criteria has been met with an acceptable degree of confidence.
 - Define verification activities. For each verification instance, define a specific verification action that will result in objective evidence needed to verify the SoI meets one or more requirements per the defined verification strategy.
 - Define verification procedure requirements for each verification action. The verification procedure requirements are requirements that will drive the formulation of steps and actions for a given verification procedure.
- – Identify constraints and objectives from the verification strategy to be incorporated within the sets of system requirements, architecture, and design. These requirements are needed to support the defined strategy.
- – Plan for the necessary enabling systems or services needed through the life cycle for verification. Enabling systems include organizational support, verification equipment, simulators, emulators, test beds, test automation tools, facilities, etc.
- – Ensure enabling system or service access needed to support verification. This includes confirming everything required for the verification activities will be available, when needed and have passed their own verification and validation. The acquisition of the

enablers can be done through several ways such as rental, procurement, development, reuse, and subcontracting.

- *Perform verification.*
 - – Define the procedures for the verification actions. A procedure can support one action or a set of actions.
 - – Execute the verification procedures for planned verification actions.
 - – Schedule the execution of verification procedures. Each scheduled verification event represents a commitment of personnel, time, resources, and equipment that would ideally show up on a project's schedule.
 - – Ensure readiness to conduct the verification procedures: availability and configuration status of the system/entity, the availability of the verification enablers, qualified personnel or operators, resources, etc.
- *Manage results of verification.*
 - – Record the verification results and any defects identified. Maintain the results in verification reports and records per organizational policy as well as contractual and regulatory requirements.
 - – Analyze the verification results against the verification success criteria to determine whether the entity being verified meets those criteria with an acceptable degree of confidence.
 - – Throughout verification, capture operational incidents and problems and track them until final resolution. Problem resolution and any subsequent changes will be handled through the Project Assessment and Control process (see [Section 2.3.4.2](#)) and the Configuration Management process (see [Section 2.3.4.5](#)). Any changes to the SoI definition (e.g., stakeholder needs, stakeholder requirements, system requirements, system architecture, system design, design characteristics, or interfaces) and associated engineering artifacts are performed within other Technical Processes.
 - – Obtain agreement from the approval authority that the verification criteria have been met to their satisfaction. Combine the individual verification records into a verification approval package for the

entity being verified and submit to the verification approval authority. The verification approval authority is the party authorized to determine whether sufficient evidence has been provided to show that the entity has passed verification with an acceptable degree of confidence.

- – Establish and sustain traceability (verification). Establish and maintain bidirectional traceability of the verified entity and verification artifacts with the system architecture and design characteristics or requirements against which the entity is being verified.
- – Give CM the information items, work products, or other artifacts needed for baselines. The Configuration Management process (see [Section 2.3.4.5](#)) is used to establish and maintain baselines. The Verification process identifies candidates for baseline and provides the items to the Configuration Management process.

Common approaches and tips.

- Identify the enabling systems and materials needed for verification early in the life cycle to allow for the necessary lead time to obtain or access them.
- Avoid conducting verification only late in the schedule or reducing the number of verification activities due to budget or schedule issues, since discrepancies and errors are more costly to correct later in the system life cycle.
- Review requirements as they are defined to ensure that the entities to which they apply can be verified against those requirements.

Elaboration

This section elaborates and provides “how-to” information on the Verification process. Additional guidance on verification can be found in the INCOSE NRM (2022) and INCOSE GtVV (2022).

Verification Planning. Planning for verification should begin when the system requirements are being defined. As the system requirements are defined, it is recommended to define the verification success criteria, method, and strategy and obtain acquirer and approval authority approval. Early planning helps drive cost and schedule estimates of the verification plan earlier in the project—maximizing the chance the full verification plan will be resourced.

Reduction of Verification Activities and Risk. If verification activities must be reduced due to cost and schedule concerns or other constraints, this should be done using a risk-based approach. The SE practitioner is urged to resist the temptation to blindly reduce the number of, or the costliest, verification activities due to budget or schedule concerns. Gaps and misses are more costly and time consuming to correct later in the life cycle—especially when these gaps show up at the integrated SoI level from reduced system element verification. If additional resources become available that allow an opportunity to verify to additional depth, the project should do so to reduce risk and increase the degree of confidence.

Notion of a Verification Action. A verification action describes verification in terms of an entity, the reference item against which the entity will be verified (e.g., a requirement, design characteristic, or standard), the expected result (success criteria deduced from the reference item the entity is being verified against), the verification strategy and method to be used, and on which level of integration of the system (e.g., system, system element). The performance of a verification action onto the submitted entity provides an obtained result which is compared with the expected result as defined by the verification success criteria. The comparison enables the determination of the acceptable conformance of the entity to the reference item with some degree of confidence. [Figure 2.50](#) illustrates several common verification actions.

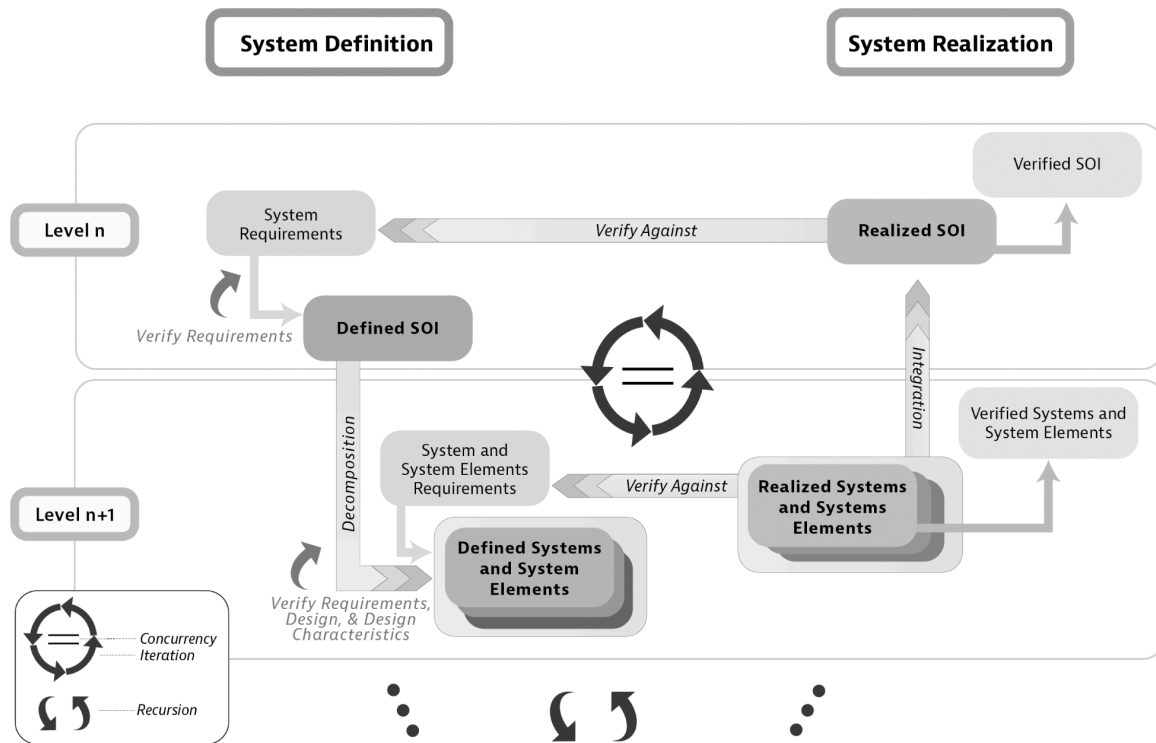


FIGURE 2.50 Verification per level. INCOSE SEH original figure created by Walden from Faisandier. Usage per the INCOSE Notices page. All other rights reserved.

Examples of verification actions include:

- *Verification of a stakeholder requirement (requirement verification)*—(1) Verify the stakeholder requirement statement correctly transforms the source or stakeholder need from which it was transformed or derived and (2) verify the stakeholder requirement satisfies the characteristics of good requirement statements (see [Section 2.3.5.3](#)).
- *Verification of a system requirement (requirement verification)*—(1) Verify the system requirement statement correctly transforms the source, stakeholder requirement, or parent from which it was transformed or derived and (2) verify the system requirement satisfies the characteristics of good requirement statements (see [Section 2.3.5.3](#)).
- *Verification of a model or simulation (model or simulation verification)*—(1) Verify that the model/simulation meets its requirements consistent with its intended purpose, (2) verify the model/simulation against syntactic and grammatical rules, characteristics, and standards defined for the type of model/simulation, and (3) verify the correct application of the appropriate patterns and heuristics used and the correct usage of modeling/simulation techniques or methods as defined by the

organization's guidelines and requirements concerning model/simulation development and use.

- *Verification of the system architecture (architecture verification)*—(1) Verify that the SoI architecture, when realized by design, will result in a SoI that will pass system verification and (2) verify the correct application of the appropriate patterns and heuristics used and the correct usage of *architecture definition* techniques or methods as defined by the organization's guidelines and requirements concerning system architecture definition.
- *Verification of the system design (design verification)*—(1) Verify that the SoI design and associated design characteristics meets its system requirements and would result in a SoI that will pass system verification with an acceptable degree of confidence and (2) verify the correct usage of patterns, trade rules, or state of the art related to the concerned technology (e.g., software, mechanics, electronics, biology, chemistry) as defined by the organization's guidelines and requirements concerning system design.
- *Verification of a realized system (product, service, or enterprise) or system element (system verification)*—Verify the system or system element meets its system requirements and design characteristics with an acceptable degree of confidence.

Verification Methods. Basic verification methods are as follows (ISO/IEC/IEEE 29148, 2018):

- **Inspection** . An examination of the item against visual or other evidence to confirm compliance with requirements. Inspection is used to verify properties best determined by examination and observation (paint color, weight, etc.). Inspection is generally non-destructive and typically includes the use of sight, hearing, smell, touch and taste; simple physical manipulation; mechanical and electrical gauging; and measurement.
- **Analysis** (*including modeling and simulation*). Use of analytical data or simulations under defined conditions to show theoretical compliance. Used where testing to realistic conditions cannot be achieved or is not

cost-effective. Analysis (including simulation) may be used when such means establish that the appropriate requirement is met by the proposed solution. Analysis may also be based on “similarity” by reviewing a similar system or system element’s prior verification and confirming that its verification status can legitimately be transferred to the present system or system element. Similarity can only be used if the systems or system elements are similar in design, manufacture, and use; equivalent or more stringent verification specifications were used for the similar system or system element; and the intended operational environment is identical to or less rigorous than the similar system or system element.

- **Demonstration** . A qualitative exhibition of functional performance, usually accomplished with no or minimal instrumentation or test equipment. Demonstration uses a set of test activities with system stimuli selected by the supplier to show that system or system element response to stimuli is suitable or to show that operators can perform their allocated functions when using the system. Often, observations are made and compared with predetermined responses.
- **Test** . An action by which the operability, supportability, or performance capability of an item is quantitatively verified when subjected to controlled conditions that are real or simulated. These verifications often use special equipment or instrumentation to obtain accurate quantitative data for analysis to determine verification.

Verification per Level. The SoI may have a number of hierarchical layers of system elements within its architecture. The planning of the verification is done recursively at each lower level as the definition of the system or a system element evolves. The execution of the verification actions occurs recursively for each layer as the elements are integrated as shown in [Figure 2.50](#). For example, the stakeholder requirements are verified to ensure they meet their higher-level requirements, the system and system element requirements are verified to ensure they meet their higher-level system requirements and the system architecture and design are verified to ensure they meet their system or system element requirements. Additionally, every layer of realized systems and system elements are verified to ensure they

meet their system requirements before being integrated into the next higher level of the SoI architecture. Any issues or discrepancies must be corrected before a system element is integrated into the next higher level of the SoI. Having passed verification at a given level, that set of elements are integrated into the next higher-level system as defined in the Integration process (see [Section 2.3.5.8](#)). System integration, system verification, and system validation continues until the integrated SoI has passed system verification.

Early Verification and MBSE. With the increased use of models and simulations as part of the design process, verification activities can be conducted earlier in the life cycle prior to implementation. Doing so will reduce the risk of issues and anomalies being discovered during system integration, system verification, and system validation activities with the actual physical hardware, mechanisms, and software and reduce the resulting expensive and time-consuming rework.

However, the SE practitioner is cautioned to resist substituting verification of the realized system with the verification results obtained using models and simulations, unless necessary. Doing so reduces the confidence level (as compared to verification against the actual realized system) and adds risk of the realized system failing system validation. As long as the realized system is not completely integrated and/or has not been validated to operate in the actual operational environment by the intended users, no result must be regarded as definitive until the acceptable degree of confidence is realized.

Managing the project's system verification program. In the progress of the project, it is important to know, at any time, the status of the verification activities, anomalies discovered, and noncompliances. This knowledge enables the project to better manage the budget and schedule as well as estimate the risks of noncompliance against the possibly of eliminating some of the planned verification actions to meet budget and schedule constraints.

2.3.5.10 Transition Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.4.10.1] The purpose of the Transition process is to establish a capability for a system to provide services specified by stakeholder requirements in the operational environment.

Description

The Transition process installs a SoI into its operational and maintenance environment. This process makes the SoI an integral part of the acquiring organization systems, business processes, and capabilities so the organization starts to benefit from using and sustaining the system's services.

The Transition process coordinates with verification and validation performed in the target environment, with the activities of operation and maintenance of new systems and services, and with the disposal of systems, system elements, materials, and services no longer needed for operation.

Transition may identify system requirements and design gaps. It may also drive changes, augmenting the initial stakeholder and system requirements.

Inputs/Outputs Inputs and outputs for the Transition process are listed in [Figure 2.51](#). Descriptions of each input and output are provided in Appendix E.

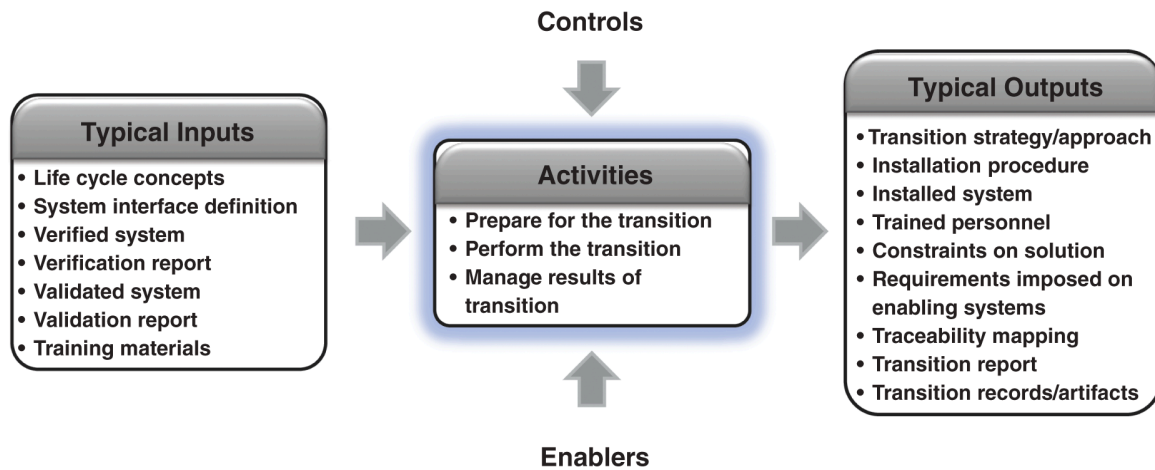


FIGURE 2.51 IPO diagram for Transition process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Transition process includes the following activities:

- *Prepare for the Transition.*
 - – Analyze the intended environment for the system deployment, including the physical sites, information technology infrastructure, organizational structure, and processes of the receiving organization.
 - – Identify the changes to the existing environment to accommodate the system.
 - – Identify and obtain (e.g., procure, develop, reuse, rent, schedule, subcontract) the requisite enabling systems, controls, products, or services required for the transition, including the changes in the environment.
 - – Plan for coordinating the development of the SoI with the modifications of its intended environment.
 - – Determine the transition team structure, composition, and responsibilities for the transition activities.
 - – Plan for the system’s transition, including allocating time and budget for all parts of the transition.
 - – Plan for mitigation strategies if the transition if the system encounters difficulties. Ensure that stakeholders understand the process/risk of possible downtime of the system and even actions to restore the predecessor system (roll-back) and the “point-of-no-return.”

- – Plan the configuration management of the system's adaptation to the local operation and support context.
- – Develop procedures for system deployment and service activation, incremental and staged if appropriate.
- – Develop procedures to validate the system and services at all relevant sites, either physical or virtual.
- – Staff, organize, and train collaborative transition teams.
- *Perform the Transition.*
 - – Deploy the system to operation, support, and maintenance sites.
 - – Invoke integration and verification processes to realize operable local system configurations.
 - – Establish systems, processes, and organizational capabilities for ongoing adaptation of the system to evolving context, including capabilities for integration with other systems, deployment to other sites, performance monitoring, and problem detection, investigation, and correction.
 - – Train the operation, maintenance, and other personnel. As applicable, perform complete review and hand-off of the operator, maintenance, and support manuals. Affirm that the personnel have the knowledge and skill levels necessary to operate, maintain, and support the system.
 - – Provide as-built information for configuration management.
 - – Activate/commission the system's services at each site. Ensure that the system delivers its intended services as expected, including collaboration with other systems and personnel.
 - – Receive final confirmation that the installed system can provide its required functions and be sustained. Assure that the system has been properly installed and verified and all issues and action items have been resolved. Assure that all agreements about developing and delivering a fully supportable system have been fully satisfied or adjudicated.
 - – Perform or support contractual acceptance of the system by the acquirer, followed by transfer of control, responsibility, ownership, and custody.
- *Manage results of Transition.*

- – Capture incidents, problems, and anomalies. Investigate and document issues. Perform corrective actions as needed. Use the Quality Assurance process for managing incidents and problem resolution. If the transition is to multiple sites using a phased approach, ensure that any corrective actions are incorporated into the transition approach.
- – Use the experience gained in the current transition instances for improving future instances.
- – Maintain bidirectional traceability of the transitioned system elements, system services, and operational capabilities with the architecture, design, and system requirements. Initiate changes as needed.

Common approaches and tips.

- Identify the enabling systems and materials needed for transition early in the life cycle to allow for the necessary lead time to obtain or access them.

Elaboration

Transition Concepts. The Transition process is not limited to the SoI going into service as a part of the operating organization. Each system element undergoes transition during its integration into a larger element, and the element's transition must be formalized in the agreements between key stakeholders, such as prime contractors and its subcontractors.

The Transition process coordinates the system or system element deployment and activation with the modification of its environment. It pays particular attention to integrating the SoI and other systems in its environment. The Transition process should be fully integrated with an organizational change process led by the receiving organization, usually incremental and staged.

The Transition process comprises all activities required to establish the capability for a system to provide services for the benefit of the

organization acquiring the system. The transition transfers the system from the development context (“system-in-the-lab”) to the utilization context (“system-in-the field”). Successful transition typically marks the beginning of the SoI or system element’s utilization stage.

Transition Considerations. The transition of new systems to a newly created organization (or a new element into a new system) differs from transitioning a new system or element into an existing organization or system. The former is sometimes referred to as “greenfield” or “clean sheet” transitioning, and the latter as “brownfield” or “legacy systems” (see Sections 4.3.1 and 4.3.2). The introduction of the new element disrupts the existing environment, so considerable effort must be invested to transition to the “new norm.”

A phase of provisional operation (also referred to as “burn-in”) is sometimes included in the transition activities, allowing operations to get used to the new system before acceptance, resulting in concurrent and iterative application of the Transition and Operation processes. Burn-in involves activities taken to operate a system element in the operational or simulated environment to detect failures and improve reliability. Usually, the operation of the system is done at levels that would cover or exceed the range of expected environmental values (heat, vibration, power, etc.). The warranty period may delay the transfer of responsibility for the system maintenance, resulting in concurrency and iteration between the Transition and Maintenance Processes.

2.3.5.11 Validation Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.4.11.1] The purpose of the Validation process is to provide objective evidence that the system, when in use, fulfills its business or mission objectives and stakeholder needs and requirements, achieving its intended use in its intended operational environment.

Description The Validation process can be applied to any engineering artifact, entity, or information item that has contributed to the definition and realization of the SoI (e.g., validation of stakeholder needs, stakeholder requirements, system requirements, models, simulations, the system architecture, design characteristics, validation procedures, or a realized system or system element). The Validation process provides objective evidence with an acceptable degree of confidence to confirm:

- 1. The “right” artifact or entity has been made according to the stakeholder needs and stakeholder requirements.
- 2. Whether or not these artifacts, entities, or information items, will result in the right SoI, when realized, that can be validated to accomplish its intended use in its operational environment when operated by its intended users.
- 3. The system does not enable unintended users to negatively impact the intended use of the system or use the system in an unintended way.

As is often stated, validation is intended to ensure that “the right artifact or entity will be or was built,” while verification is intended to ensure that “the artifact or entity has been built right.”

Inputs/Outputs Inputs and outputs for the Validation process are listed in [Figure 2.52](#). Descriptions of each input and output are provided in Appendix E.

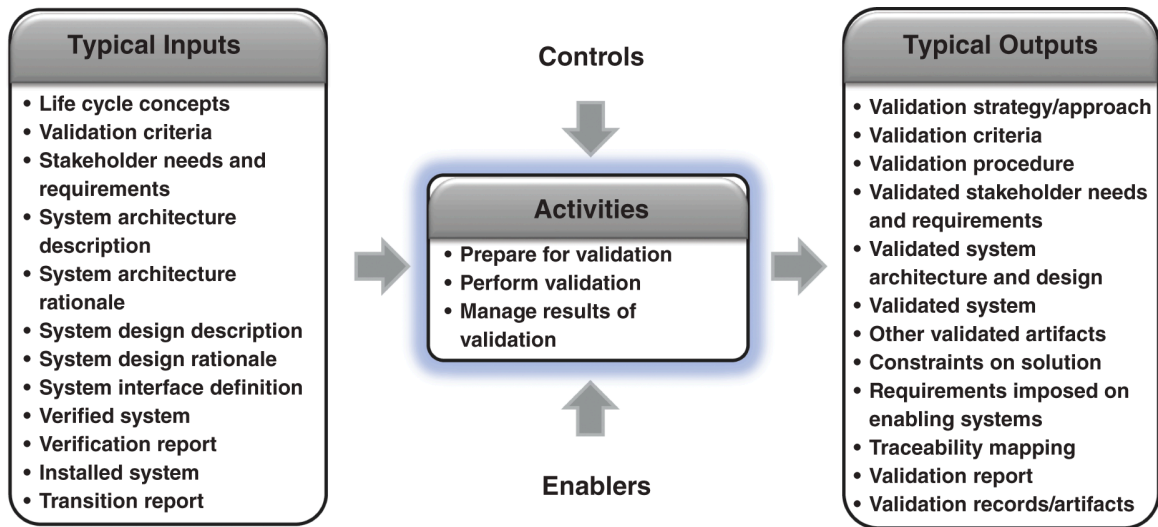


FIGURE 2.52 IPO diagram for Validation process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Validation process includes the following activities:

- *Prepare for validation.*
 - – Define the scope (what will be validated) and the validation actions (strategy, method, and success criteria). Validation activities consume resources: time, labor, facilities, and funds. The scope of the organization’s validation strategy/approach should be documented within the project’s SEMP and system integration, verification, and validation plans.
 - Establish a list of artifacts, entities, or information items to be validated.
 - Identify the stakeholder needs and stakeholder requirements against which each entity will be validated.
 - – Consider and capture constraints that could impact the feasibility or effectiveness of validation actions. The constraints could impact the implementation of the validation actions and include contractual constraints, limitations due to regulatory requirements, cost, schedule, feasibility to exercise a function, safety and security considerations, the laws of physics, physical configurations, accessibility, etc.
 - – For each validation action, select one or more validation methods and associated success criteria. Validation methods are similar to the methods defined for verification (inspection, analysis,

demonstration, or test) (see [Section 2.3.5.9](#)). The success criteria define what the validation actions must do that will result in sufficient objective evidence to show that the entity has fulfilled the need(s) or requirement(s) against which it was validated against, achieving its intended use in its intended operational environment by its intended users.

- – Establish the strategy/approach for validation, including trade-offs between scope and constraints. The validation strategy includes the method that will result in objective evidence that the validation success criteria have been met with an acceptable degree of confidence. Significant collaboration is necessary with the stakeholders and approval authority to ensure there is agreement on what is necessary to accept the validation results.
 - Define validation activities. For each validation instance, define a specific validation action that will result in objective evidence needed to validate the SoI meets one or more stakeholder needs or stakeholder requirements per the defined validation strategy.
 - Define validation procedure requirements for each validation action. The validation procedure requirements are requirements that will drive the formulation of steps and actions for a given validation procedure.
- – Identify constraints and objectives from the validation strategy to be incorporated within the sets of stakeholder needs and requirements and the system requirements transformed from them.
- – Plan for the necessary enabling systems or services needed through the life cycle for validation. Enabling systems include organizational support, validation equipment, simulators, emulators, test beds, test automation tools, facilities, etc.
- – Ensure enabling system or service access needed to support validation. This includes confirming everything required for the validation activities will be available, when needed. The acquisition of the enablers can be done through several ways such as rental, procurement, development, reuse, and subcontracting.
- *Perform validation.*

- – Define the procedures for the validation actions. A procedure can support one action or a set of actions.
- – Execute the validation procedures for planned validation actions.
 - Schedule the execution of validation procedures. Each scheduled validation event represents a commitment of personnel, time, resources, and equipment that would ideally show up on a project's schedule. At the integrated SoI level, this should be done against the actual SoI in the operational environment or one as close to it as possible, by the intended users or equivalent surrogates.
 - Ensure readiness to conduct the validation procedure: availability and configuration status of the system/entity, the availability of the validation enablers, qualified personnel or operators, resources, etc. At the integrated SoI level, since it often depends on customer and intended user involvement, this can be particularly important to plan out in advance to be sure the right individuals are present.
- – *Manage results of validation.*
 - – Record the validation results and any defects identified. Maintain the results in validation reports and records per organizational policy as well as contractual and regulatory requirements.
 - Analyze the validation results against the validation success criteria to determine whether the entity being validated meets those criteria with an acceptable degree of confidence.
 - – Throughout validation, capture operational incidents and problems and track them until final resolution. Problem resolution and any subsequent changes will be handled through the Project Assessment and Control process (see [Section 2.3.4.2](#)) and the Configuration Management process (see [Section 2.3.4.5](#)). Any changes to the SoI definition (e.g., stakeholder needs, stakeholder requirements, system requirements, system architecture, system design, design characteristics, or interfaces) and associated engineering artifacts are performed within other Technical Processes
 - – Obtain agreement from the approval authority that the validation criteria have been met to their satisfaction. Combine the individual

validation records into a validation approval package for the entity being validated and submit to the validation approval authority. The validation approval authority is the party authorized to determine whether sufficient evidence has been provided to show that the entity has passed validation with an acceptable degree of confidence.

- – At the integrated SoI level, validation may be performed with or by the acquirer as defined in the supplier agreement. However, at lower levels in the architecture, validation may be performed by the supplier without acquirer direct involvement.
- – Establish and sustain traceability (validation). Establish and maintain bidirectional traceability of the validated entity and validation artifacts with the system architecture, system design, models, and the stakeholder needs and stakeholder requirements against which the entity is being validated.
- – Give CM the information items, work products, or other artifacts needed for baselines. The Configuration Management process (see [Section 2.3.4.5](#)) is used to establish and maintain configuration items and baselines. The validation process identifies candidates for baseline, and then provides the items to the Configuration Management process.

Common approaches and tips.

- Identify the enabling systems and materials needed for validation early in the life cycle to allow for the necessary lead time to obtain or access them.
- Validation also reveals the effects the SoI may have on enabling, interfacing, and interoperating systems. Validation actions and analysis should include these system interactions in the scope.
- Involve the broadest range of stakeholders that is practical, including end users and operators,
- Validation should include actions that provide insight as early as possible, such as analysis, modeling, and simulation of anticipated operational characteristics and system behavior.

- Start to develop the validation planning as the OpsCon, operational scenarios, stakeholder needs, and stakeholder requirements are defined. Early consideration of the potential validation actions and methods helps to anticipate constraints, costs, and necessary enablers, as well as start the acquisition of those enablers.
- Validation actions during the Business or Mission Analysis process (see [Section 2.3.5.1](#)) include assessment of the OpsCon through operational scenarios that exercise all system operational modes and demonstrating system-level performance.

Elaboration

This section elaborates and provides “how-to” information on the Validation process. Additional guidance on validation can be found in the INCOSE NRM (2022) and INCOSE GtVV (2022).

General Considerations. The stakeholder needs and stakeholder requirements the SoI is being validated against are derived from the mission statement, goals, objectives, critical measures, constraints, risks, and set of life cycle concepts for the SoI defined by the organization or acquirer during the Stakeholder Needs and Requirements Definition and System Requirements Definition processes (see Sections 2.3.5.2 and 2.3.5.3). The life cycle concepts include scenarios and use cases that are performed in a specific operational environment by the intended users for not only operation, but during other life cycle stages including production, operation, support, and retirement. It is common for these scenarios and use cases to be exercised during the conduct of the validation procedures within the operational environment with the intended users. The common saying “test as you fly, fly as you test” applies. When using scenarios and use cases, in addition to nominal operations, it is important to also address off-nominal, alternate cases, misuse cases, and loss scenarios. A positive validation result obtained in a given environment by specific users can turn noncompliant if the environment or class of users change. These changes may not be immediately known by the developer; however,

changing stakeholder needs and stakeholder requirements should be accommodated by the acquirer and developer's SE processes.

During validation, especially for walkthroughs and similar activities, it is highly recommended to involve intended users/operators. Validation will often involve going back directly to the users to have them perform an acceptance test under their own local operational conditions in the intended operational environment. When the system is validated at a supplier facility or organization, the acquirer will often want to conduct additional validation activities in their own facility, in the intended operational environment, and with the intended users. The stakeholders who were involved in defining the life cycle concepts and needs must be presented with the results of the validation activities to ensure their needs and requirements have been met.

Validation Planning. Planning for validation should begin when the stakeholder needs and stakeholder requirements are being defined. As they are defined, it is recommended to define the validation success criteria, method, and strategy and obtain acquirer and approval authority approval. Early planning helps drive cost and schedule estimates of the system validation plan earlier in the project—maximizing the chance the full system validation plan will be resourced.

Reduction of Validation Activities and Risk. If validation activities must be reduced due to cost and schedule concerns, this should be done using a risk-based approach. The SE practitioner is urged to resist the temptation to blindly reduce the number of, or the costliest, validation activities due to budget or schedule concerns. Gaps and misses are more costly and time consuming to correct later in the life cycle—especially when these gaps show up at final system acceptance by the acquirer or regulatory agency. If additional resources become available that allow an opportunity to validate lower-risk, non-critical stakeholder needs and stakeholder requirements, the project should do so to reduce risk and increase the degree of confidence.

Notion of a Validation Action. Validation actions are similar to verification actions, and the reader is referred to the Verification process (see [Section 2.3.5.9](#)) for background. [Figure 2.53](#) illustrates several common validation actions.

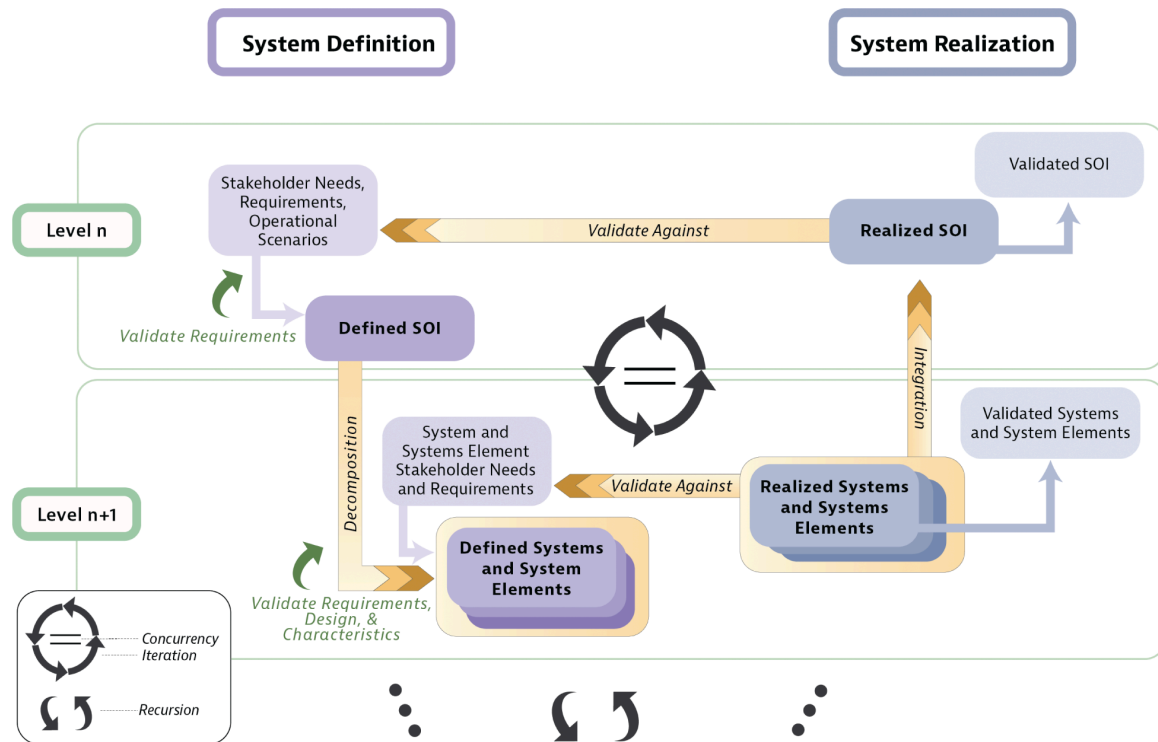


FIGURE 2.53 Validation per level. INCOSE SEH original figure created by Walden from Faisandier. Usage per the INCOSE Notices page. All other rights reserved.

Examples of validation actions include:

- *Validation of a stakeholder requirement (requirement validation)*—Validate that the stakeholder requirement is the right requirement and clearly and accurately communicates the need of the stakeholder, is in the stakeholder’s language, and is actionable (i.e., can be transformed into one or more system requirements). For stakeholder requirements and sets of stakeholder requirements ask, “If a SoI were built to these requirements, would the SoI meet the needs from which these requirements were transformed?”
- *Validation of a system requirement (requirement validation)*—Validate that the system requirement is the right requirement and clearly and accurately communicates the need and requirement of the stakeholder, is expressed in technical terms, and is actionable (i.e., can be

transformed into a system architecture and design). For system requirements and sets of system requirements ask, “If a SoI were built to the system architecture and design transformed from these requirements, would the SoI meet the intent of the requirements from which the architecture and design were transformed?”

- *Validation of a model or simulation (model or simulation validation)*—(1) Validate that the model/simulation accurately reflects the intended behavior of the entity it represents in its operational environment when operated by the intended users and (2) validate that the model/simulation meets the intended purpose for which it was developed.
- *Validation of the system architecture (architecture validation)*—Validate that the architecture is the right architecture that will result in a design for the SoI that will meet the stakeholder needs and stakeholder requirements.
- *Validation of the system design (design validation)*—Validate that the design, as communicated by the design characteristics, will result in a SoI that meets its intended purpose in its operational environment when operated by the intended users as defined by the stakeholder needs and stakeholder requirements.
- *Validation of a realized SoI (product, service, or enterprise) (system validation)*—Validate that realized SoI meets its intended purpose in its operational environment when operated by the intended users and does not enable unintended users to negatively impact the intended use of the system or use the system in an unintended way with an acceptable degree of confidence as defined by the stakeholder needs and stakeholder requirements.

Validation Outcomes. Typical validation outcomes include:

- **Acceptance.** Acceptance is an activity conducted prior to transition to the acquirer such that the acquirer can decide if this transition is appropriate. A set of operational validation actions is often exercised, or a review of validation results performed by the supplier is systematically performed as part of acceptance.

- **Certification.** Certification is a written assurance that the system has been developed per a defined procedure and can perform its intended functions in accordance with identified legal or industrial standards (e.g., airworthiness standards for aircraft, information assurance). A host of information can be part of the certification package, including development reviews, verification results, and validation results. However, certification is typically performed by outside authorities, without direction as to how the needs are to be validated. For example, this method is used for electronics devices via Conformité Européenne (CE) certification in Europe and via Underwriters Laboratories (UL) certification in the United States and Canada.
- **Readiness for Use.** As part of the analysis of the validation results, the project team and validation authority may need to make a readiness for use assessment. This may occur several times in the life cycle, including upon first article delivery, upon completion of production (if more than a single system is produced), following maintenance actions, or successful completion of field trials with a predefined user population. In the field, particularly after maintenance, it may be necessary to establish whether the system is ready for reintroduction to service.
- **Qualification.** System qualification requires that all verification and validation actions have been successfully performed, documented, and that the SoI is “qualified” for use as intended by the supplier organization. These verification and validation actions cover not only the SoI itself but also all the interfaces with its environment (e.g., for a space system, the validation of the interface between space segment and ground segment). The qualification process must demonstrate that the characteristics or properties of the realized system, including margins, meet the applicable system requirements and/or stakeholder requirements. The qualification is concluded by an acceptance review and/or an operational readiness review.

Validation per Level. The SoI may have a number of hierarchical layers of system elements within its architecture. The planning of the validation is done recursively for each level as the definition of the system or a system element evolves. The execution of the validation actions occurs recursively

for each layer as the elements are integrated as shown in Figure 4.53. For example, the stakeholder needs and stakeholder requirements are validated against the stakeholder real world expectations to ensure they are the right stakeholder needs and stakeholder requirements, the systems requirements are validated against the stakeholder needs and requirements to ensure they are right system requirements, and the system architecture and design are validated against the stakeholder needs and requirements to ensure they are the right system architecture and design. Additionally, every layer of realized systems and system elements are validated to ensure they meet their stakeholder needs and stakeholder requirements in their operational environment before being integrated into the next higher level of the SoI architecture. Having passed system verification and system validation at a given level, that system element is integrated into the next higher-level system as defined in the Integration process (see [Section 2.3.5.8](#)). System integration, system verification, and system validation continue until the integrated SoI has passed system validation.

Early System Validation and MBSE. With the increased use of models and simulations as part of the design process, validation activities can be conducted earlier in the life cycle prior to implementation. Doing so will reduce the risk of issues and anomalies being discovered during system integration, system verification and system validation activities with the actual physical hardware, mechanisms, and software and reduce the resulting expensive and time-consuming rework.

In addition, modeling and simulations early in the project allows not only expectation management but also early feedback from the acquirer and other stakeholders on the final system architecture and design before implementation. It will be much less expensive and time consuming to resolve issues before the realization of the actual physical hardware and software and before system integration, system verification, and system validation activities.

Because the behavior of a system is a function of the interaction of its elements, a major goal of systems validation is assessing the behavior of the integrated physical system and identifying emergent properties not specifically addressed in the stakeholder needs or stakeholder requirements nor identified during modeling and simulations. Emergent properties may be positive or negative. For example, cascading failures across multiple interface boundaries between the system elements that are part of the SoI's architecture. Relying on models and simulations of the SoI and operational environment may not uncover all the emerging properties and issues that occur in the physical realm. While validation using models and simulations allows a theoretical determination that the modeled system will meet its needs in the operational environment by the intended users once realized, the assessment of the actual system behavior (system validation) must be done, whenever possible, in the physical realm with the actual hardware and software integrated into the higher-level system which it is a part in the actual operational environment by the intended users.

There are cases when it may not be practical in terms of the intended use and actual operational environment to do all system validation activities. However, the SE practitioner is cautioned to not substitute validation of the realized system with the validation results obtained using models and simulations, unless absolutely necessary. Doing so adds risk to the project and reduces the confidence level (as compared to validation against the actual realized system in its actual operational environment when operated by the intended users) and adds risk of the realized system failing system validation when delivered to the acquirer or submitted to a regulatory agency. As long as the realized system is not completely integrated and/or has not been validated to operate in the actual operational environment by the intended users, no result must be regarded as definitive until the acceptable degree of confidence is realized.

Managing the project's validation program. In the progress of the project, it is important to know, at any time, the status of the validation activities, anomalies discovered, and noncompliances. This knowledge enables the

project to better manage the budget and schedule as well as estimate the risks of noncompliance against the possibility of eliminating some of the planned validation actions to meet budget and schedule constraints.

2.3.5.12 Operation Process

Overview

Purpose

As stated in ISO/IEC/IEEE 15288,

[6.4.12.1] The purpose of the Operation process is to use the system to deliver its services.

Description

The Operation process focuses on delivering services provided by the system for the benefit of the operating organization. This process is often concurrent with the Maintenance process of sustaining the system's services. During Operation, the SoI functions as an integral part of the operating organization. The SoI contributes to the Business or Mission Analysis process by cooperating with human operators and diverse interfacing systems.

Operation may identify the system requirements and design gaps. It may also drive changes, augmenting the initial stakeholder and system requirements.

Inputs/Outputs Inputs and outputs for the operation process are listed in [Figure 2.54](#). Descriptions of each input and output are provided in Appendix E.

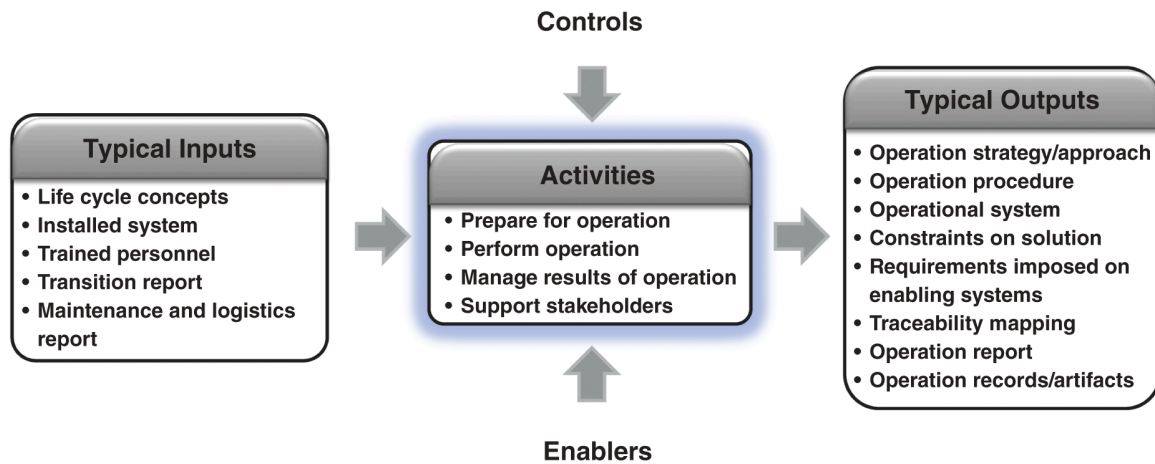


FIGURE 2.54 IPO diagram for Operation process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Operation process includes the following activities:

- *Prepare for operation.*
 - – Influence the Concept of Operations (ConOps) of the receiving organization, the Operational Concept (OpsCon) of the SoI, the stakeholder needs and requirements, and the system requirements impacting the operation of the SoI.
 - – Identify relevant regulations, legal requirements, environmental and ethical constraints.
 - – Define business rules related to modifications that sustain existing or enhanced services.
 - – Plan for operational capability build-up, including confirmation of site deployment schedules, personnel availability, training, and logistic support availability.
 - – Identify and obtain (procure, develop, reuse, rent, schedule, subcontract) the requisite enabling systems, controls, products, or services required for the operation.
 - – Verify that the SoI is accompanied by all relevant information products, such as documentation, manuals, and procedures. Identify gaps and initiate changes as necessary.
 - – Review the transition, validation, and maintenance strategies for compatibility with the OpsCon and their completeness concerning the expected operational capabilities.
- *Perform operation.*

- – Confirm completion of the system transition at the operational sites.
- – Prepare and verify the system's configurations for delivering specific services or missions.
- – Operate the system according to the established procedures. Update the procedures as experience accumulates.
- – Ensure the flow of materials, energy, and information into and from the SoI. Monitor the functioning of the systems providing inputs for the SoI and utilizing its outputs.
- – Track system performance, including operational availability. Identify, investigate, and correct problems and anomalies.
- – When abnormal operational conditions warrant, conduct planned contingency actions. Perform system contingency operations, if necessary.
- *Manage results of operation.*
 - – Capture incidents, problems, and anomalies. Investigate and document the issues. Perform corrective actions as needed. Use the Quality Assurance process for managing incidents and problem resolution.
 - – Use the experience gained during the operation for improvement.
 - – Maintain bidirectional traceability of the system's assets, services, and operational capabilities with system architecture, design, and system requirements. Initiate changes as needed.
- *Support stakeholders*
 - – While the customer is responsible for the Operation process, the supplier should support the customer throughout the system life cycle leveraging the knowledge generated by the customer and the supplier.

Common approaches and tips.

- Identify the enabling systems, products, services, and materials needed for operation early in the life cycle to allow for the necessary lead time to obtain or access them.

Elaboration

Operation Concepts. Successful operation of the SoI as a part of the operating organization is the ultimate goal of SE. The stakeholders' needs and requirements regarding operation constitute a significant source of the system requirements and a significant input to the Validation and Transition processes.

During operation, the SoI interfaces with other systems in its environment (see [Section 1.3.3](#)). These systems are SoIs in their own right, and their life cycles must be coordinated with the life cycle of your SoI.

The operational environment may change and evolve while the system is being developed. Considerable effort must be invested in recognizing these changes and updating the life cycle concepts (especially ConOps and OpsCon) and all derived requirements.

2.3.5.13 Maintenance Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.4.13] The purpose of the Maintenance process is to sustain the capability of the system to provide a service.

Description

The Maintenance process focuses on sustaining the system's ability to provide services for the operating organization's benefit. This process is often concurrent with the Operation process of delivering the system's services. Maintenance includes the activities to provide operations support, logistics, and material management to sustain satisfactory quality, performance, and availability of the system's services.

Maintenance may identify requirements and design gaps. It may also drive changes in the SoI, augmenting the initial stakeholder and system requirements.

Inputs/Outputs Inputs and outputs for the Maintenance process are listed in [Figure 2.55](#). Descriptions of each input and output are provided in Appendix E.

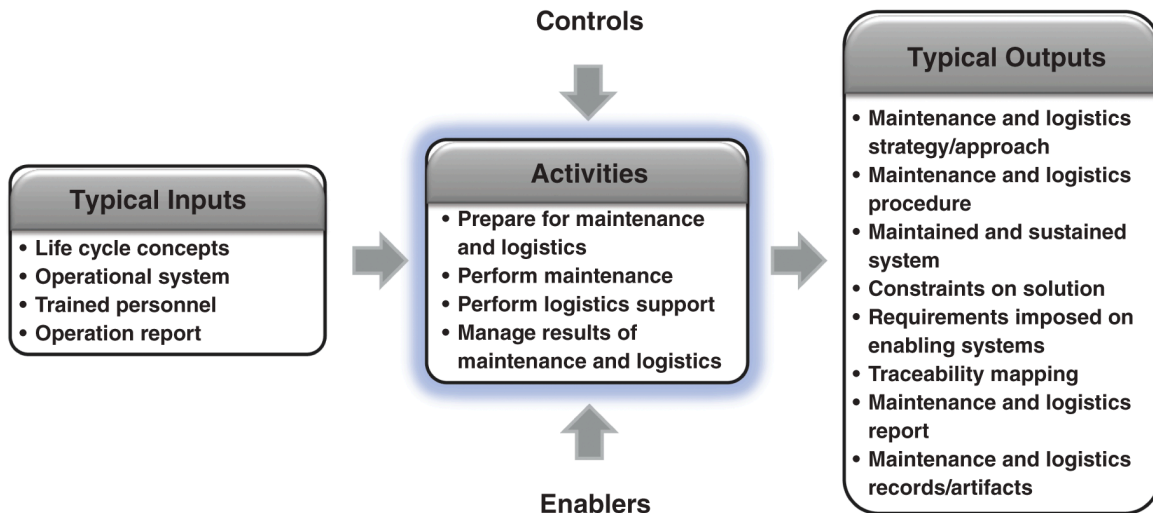


FIGURE 2.55 IPO diagram for Maintenance process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Maintenance process includes the following activities:

- *Prepare for maintenance and logistics.*
 - – Define and maintain the maintenance and logistics strategies of the SoI and its elements and update the system requirements and attribute specifications impacting the maintenance and logistics support.
 - – Define business rules related to modifications that sustain existing or enhanced services.
 - – Identify relevant regulations, legal requirements, and ethical constraints and generate corresponding requirements.
 - – Plan for maintenance and logistics support capability build-up, including site deployment schedules, personnel availability, and training, including the logistic support availability.

- – Establish appropriate warranty and licenses (e.g., software, legal) and the lines of communication to activate more support when needed.
- – Identify and obtain (procure, develop, reuse, rent, schedule, subcontract) the requisite enabling systems, controls, products, or services required for maintenance and logistics support.
- – Review the transition, validation, and operation strategies for compatibility with the support concept and their completeness concerning the expected maintenance and logistics support capabilities.
- *Perform maintenance.*
 - – Confirm completion of the system transition at the maintenance sites.
 - – Maintain the system according to the established procedures. Update the procedures when experience accumulates.
 - – Detect, identify, and repair physical and logical damage to the system, including data corruption or inaccessibility. Identify and replace faulty or obsolete parts, including software updates.
 - – Monitor the SoI and its environment to detect or predict system failures or performance degradation, identifying and resolving operational problems minimizing operational interruptions.
 - – Prevent operation disruptions by scheduling repairs and replacements before failures occur, based on operations history or failure prediction.
 - – Ensure availability of materials and parts for replacement and repairs by production, acquisition, or repairs, including operations and maintenance of logistics processes and systems. Conduct logistics operations according to the established procedures. Update the procedures when experience accumulates.
 - – Track all maintenance repairs for analysis, which may lead to performance trends that can trigger warranty claims or new project needs.
- *Perform logistic support.*
 - – Conduct acquisition logistics actions
 - – Conduct operational logistics actions

- *Manage results of maintenance and logistics.*
 - – Capture incidents, problems, and anomalies. Investigate and document the issues. Perform corrective actions as needed. Use the Quality Assurance process for managing incidents and problem resolution.
 - – Use the experience gained while performing maintenance for improvement.
 - – Maintain bidirectional traceability of the maintenance and logistics assets, services, and capabilities with system architecture, design, and system requirements. Initiate changes as needed.
 - – Manage the configuration data items.

Common approaches and tips.

- Identify the enabling systems, products, services, and materials needed for maintenance and logistics support early in the life cycle to allow for the necessary lead time to obtain or access them.
- The maintenance of the SoI must be coordinated with the maintenance of other systems in its environment (the interoperating and enabling systems). The failure or malfunction of any system can trigger maintenance actions in other systems due to technical, organizational, economic, or political concerns.

Elaboration

The Maintenance process supports the operation of the SoI and its elements throughout its life cycle. The maintenance and logistics activities regarding the SoI must be integrated into the operating organization's existing support and logistics networks. This includes provisions for sustaining the skills and competencies of personnel performing operation and maintenance.

Different modes of maintenance should be considered:

- Corrective maintenance restores system services to normal operations (e.g., remove and replace hardware, reload software, apply a software

patch).

- Preventive maintenance prevents failures and malfunctions by scheduling routine maintenance actions to sustain optimal system operational performance.
- Predictive maintenance is a more advanced preventive maintenance that utilizes data collected during the system operations to predict failures and malfunctions and schedule the maintenance actions in advance.
- System modification is a form of maintenance that extends the system's useful life by changing the system to sustain existing capabilities in the changing environment. Adding new capabilities (system upgrades) is sometimes considered part of the maintenance.

2.3.5.14 Disposal Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.4.14.1] The purpose of the Disposal process is to end the existence of a system element or system for a specified intended use, appropriately handle replaced or retired elements, appropriately handle any waste products, and to properly attend to identified critical disposal needs.

The Disposal process is conducted in accordance with applicable guidance, policy, regulations, and statutes throughout the system life cycle.

Description The Disposal process generates requirements and constraints that must be balanced with defined stakeholders' needs and requirements and other design considerations. Further, environmental concerns drive the designer to consider reclaiming the materials or recycling them into new systems. Incremental disposal can be applied at any point in the life cycle (e.g., prototypes that are not to be reused or evolved, waste materials during manufacturing, parts that are replaced during maintenance). The

Disposal process may also be used to manage the transition of system elements from a current SoI to a different system.

The Disposal process also includes any steps necessary to return the environment to an acceptable condition; handle all system elements and waste products in an environmentally sound manner in accordance with applicable legislation, organizational constraints, and stakeholder agreements; and document and retain records of disposal activities, as required for monitoring by external oversight or regulatory agencies.

Inputs/Outputs Inputs and outputs for the Disposal process are listed in [Figure 2.56](#). Descriptions of each input and output are provided in Appendix E.

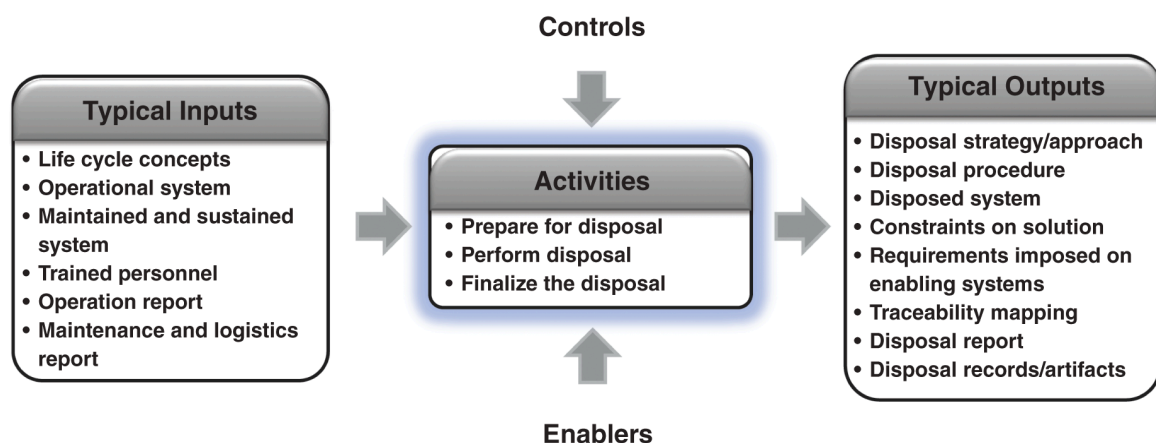


FIGURE 2.56 IPO diagram for Disposal process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Disposal process includes the following activities:

- *Prepare for disposal.*
 - – Review the retirement concept (may be called a disposal concept), including any hazardous materials and other environmental impacts to be encountered during disposal.
 - – Plan for disposal, including the development of the strategy.
 - – Impose associated constraints on the system requirements.
 - – Ensure that the necessary enabling systems, products, or services required for disposal are available, when needed. The planning includes the identification of requirements and interfaces for the

enablers. The acquisition of the enablers can be done through various ways such as rental, procurement, development, reuse, and subcontracting. An enabler may be a complete enabling system developed as a separate project from the project of the SoI.

- – Identify elements that can be reused and that cannot be reused. Special methods may need to be implemented for hazardous materials.
- – Specify containment facilities, storage locations, inspection criteria, and storage periods, if the system is to be stored.
- *Perform disposal.*
 - – Decommission the system or system elements to be disposed.
 - – Disassemble the elements for ease of handling. Include identification and processing of reusable elements.
 - – Extract all elements and waste materials that are no longer needed—this includes removing materials from storage sites, consigning the elements and waste products for destruction or permanent storage, and ensuring that the waste products or elements not intended for reuse cannot get back into the supply chain.
 - – Dispose of deactivated system elements per the disposal procedure.
 - – Ensure the disposal staff adheres to safety, security, privacy and environment regulations or policies and capture their tacit knowledge for future needs.
- *Finalize the disposal.*
 - – Confirm no adverse effects from the disposal activities and return the environment to its original state.
 - – Maintain documentation of all disposal activities and residual hazards.

Common approaches and tips:

- Consider donating an obsolete system—Many items, both systems and information, of cultural and historical value have been lost to posterity because museums and conservatories were not considered as an option during the retirement stage.

- Concepts such as zero footprint and zero emissions drive current trends toward corporate social responsibility that influence decision making regarding cleaner production and operational environments and eventual disposal of depleted materials and systems.
- Design the SoI to support the circular economy (see [Section 3.1.10](#)). Maintaining materials in closed loops maximizes material value without damaging ecosystems (McDonough, 2013).

Elaboration

The project team conducts analyses to develop solutions for disposition of the system, system elements, and waste products based on evaluation of alternative disposal methods. Methods addressed should include storing, dismantling, reusing, recycling, reprocessing, and destroying systems, system elements, materials, consumables, waste, and enabling systems,.

Disposal analyses are essential to ensure the planning and feasibility of disposal throughout the life cycle. The following are key points with respect to the analyses.

- Analyses include consideration of costs (including LCC), disposal sites, environmental impacts, health and safety issues, responsible agencies, handling and shipping, supporting items, and applicable international, national, and local regulations.
- Analyses support selection of system elements and materials that will be used in the system design and should be readdressed to consider design and project impacts from changing laws and regulations throughout the life cycle.

Disposal strategy and design considerations are updated throughout the system life cycle in response to changes in applicable laws, regulations, and policy.

The ISO 14000 (2015) series includes standards for environmental management systems and life cycle assessment.

3

LIFE CYCLE ANALYSES AND METHODS

3.1 QUALITY CHARACTERISTICS AND APPROACHES

3.1.1 Introduction to Quality Characteristics

ISO/IEC/IEEE 15288 (2023), Section 3.36 defines *Quality Characteristic* (QC) as: *inherent characteristic of a product, process, or system related to a requirement*. QCs are how the stakeholders will judge the quality of the system. Approaches exist that help ensure these characteristics are present in the SoI and its broader context or environment.

The objective of the following sections is to give enough information to a Systems Engineering (SE) practitioner to appreciate the significance of various QC approaches, even if they are not an expert in the subject. In previous editions of the handbook, the QC approaches were known as Specialty Engineering or the Engineering Specialties. These approaches are also known as Design for X (DFX) and Through-Life Considerations. The QCs are informally known as the -ilities since many, but not all, end in “ility” in the English language.

QC approaches, as used in this handbook, are life cycle perspectives that need to be considered to ensure the system is developed and its ecosystem

cultivated so that QCs are present when the system is produced, utilized, supported, and ultimately retired. QC approaches often generate non-functional requirements. Some QC approaches, such as safety, security, and resilience may also generate functional requirements. These QC approaches are applied throughout the system’s life cycle, as notionally shown in [Figure 3.1](#). Consideration beyond the engineered system, including the system, SoS, or enterprise that it is a part of, and its interoperating and enabling systems, is also necessary.

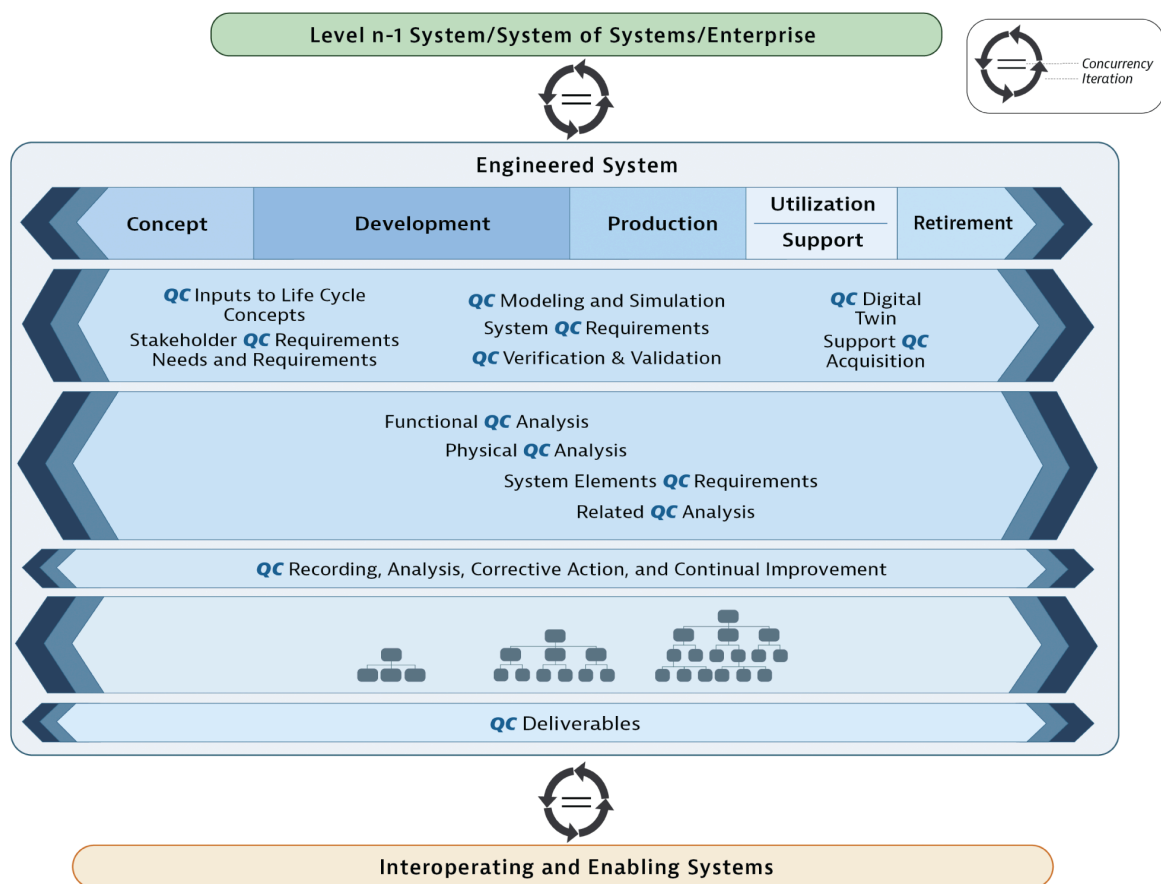


FIGURE 3.1 Quality characteristic approaches across the life cycle. INCOSE SEh original figure created by Taljaard, Kemp, and Walden. Usage per the INCOSE Notices page. All other rights reserved.

The QC approaches in this section are covered in alphabetical order by name to avoid giving more weight to one over another. [Table 3.1](#) summarizes the QC approaches included in the handbook. Not every QC approach will be applicable to every system or every application domain. It is recommended that subject matter experts are consulted and assigned as appropriate to conduct QC approaches. More information about the QC approaches can be found in references to external sources.

TABLE 3.1 Quality Characteristic approaches

QC approach	An approach that ...	Representative QCs
Affordability Analysis	maximizes value, providing cost effective capability over the entire life cycle	Affordability, Cost-Effectiveness, Life Cycle Cost (LCC), Value Robustness
Agility Engineering	enables change in a timely and cost-effective manner	Adaptability, Agility, Changeability, Evolvability, Extensibility, Flexibility, Modularity, Reconfigurability, Scalability
Human Systems Integration	integrates technology, organizations, and people effectively	Desirability, Ergonomics, Habitability, Human Factors, Human-Computer Interaction (HCI), Human-Machine Interface (HMI), Usability, User Interface (UI).